

Half Adder Building Blocks to Full Adder:

Two four bit numbers getting added:

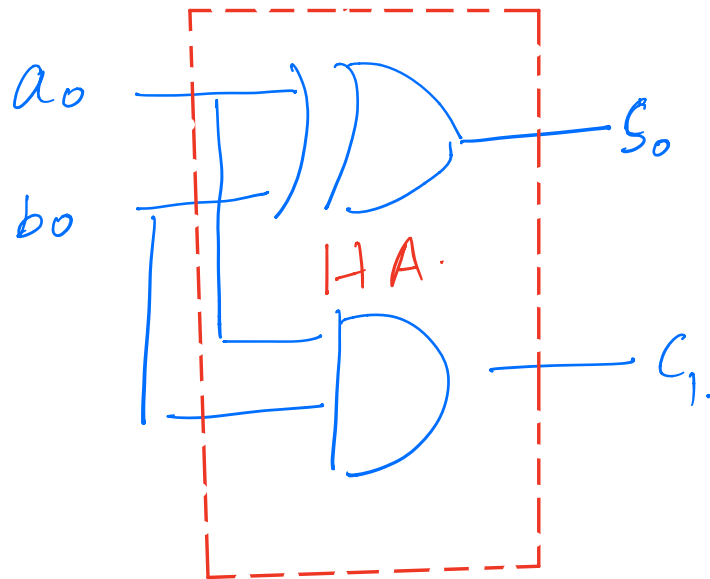
$$\begin{array}{rcccc} A_0 & = & a_3 & a_2 & a_1 & a_0 \\ B_0 & = & b_3 & b_2 & b_1 & b_0 \\ \hline & & s_3 & s_2 & s_1 & s_0 \\ & & c_4 & c_3 & c_2 & c_1 \end{array}$$

For example:

$$\begin{array}{rcccc} A_0 & 0 & 0 & 1 & 1 \\ B_0 & 1 & 0 & 0 & 1 \\ \hline \text{Sum:} & & & & \\ (S_i) & 1 & 1 & 0 & 0 \\ \text{Carry:} & & & & \\ (C_{i+1}) & 0 & 0 & 1 & 1 \\ \hline \end{array}$$

We want eventually a circuit for the above.

We have seen already:



Full Adder:

$$\begin{aligned}
 S_1 &= S_0 \oplus b_1 \\
 &= S_0 \oplus (a_1 \oplus C_1) \\
 &= S_0 \oplus a_1 \oplus C_1
 \end{aligned}$$

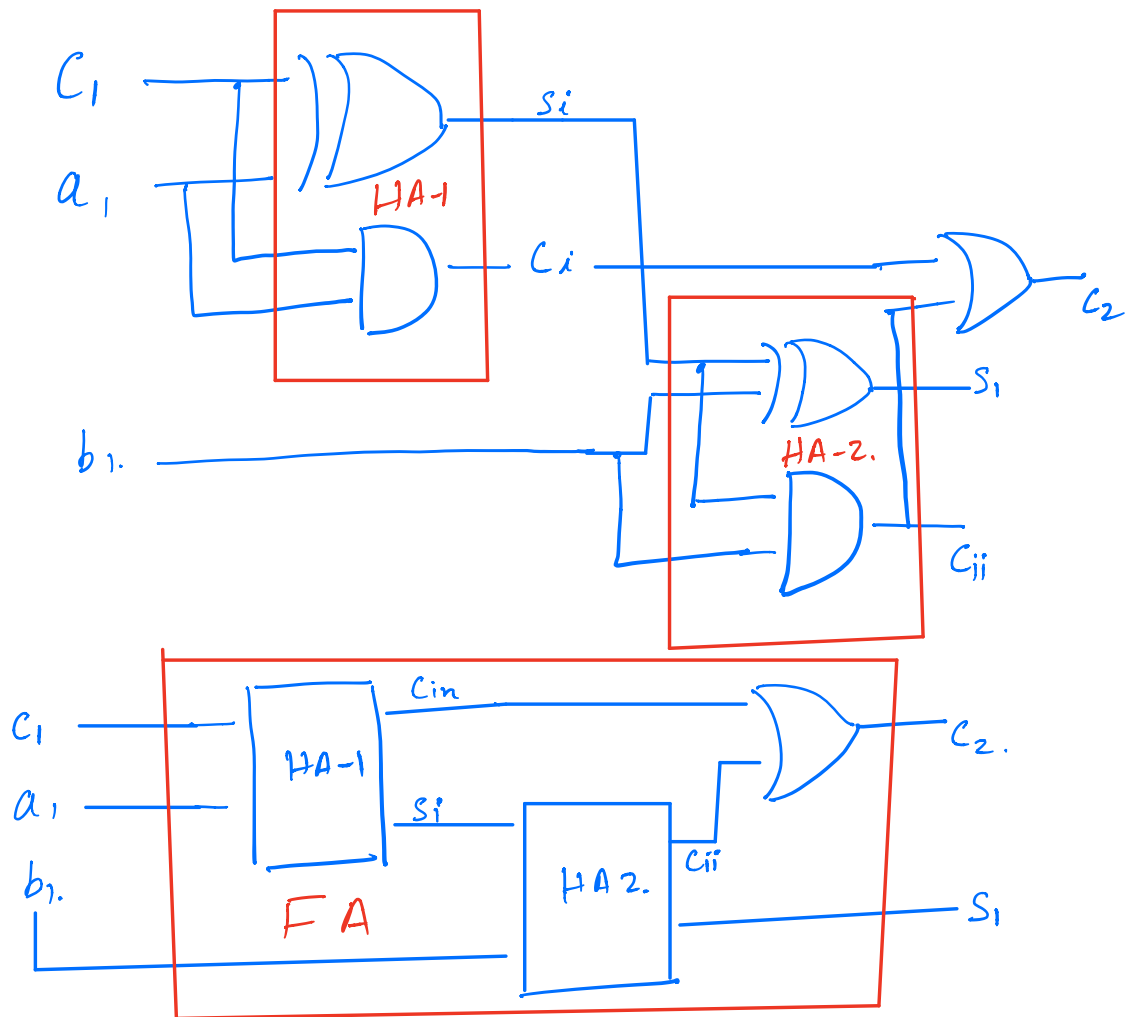
$$\begin{array}{r}
 C_1 \\
 a_1 \\
 \hline
 C_{in} \quad S_0 \\
 b_1 \\
 \hline
 C_2 \quad S_1 \\
 \hline
 \end{array}$$

C_2 is 1
 if C_{in} is 1
 OR if
 $S_0 = b_1 = 1$

$$\therefore C_2 = C_{in} + S_i b_1$$

Now $S_i b_1$ is also the HA carry for the 2nd HA. Let us call it C_{ii}

$$\underline{\text{Then } C_2 = C_{in} + C_{ii}}$$



Now prove from the truth table below

$$S_1 = C_1 \oplus a_1 \oplus b_1$$

$$C_2 = C_{in} + C_{11} = C_1 a_1 + S_1 b_1 = C_1 a_1 + (C_1 \oplus a_1) b_1 \quad \rightarrow (1)$$

C_1	a_1	b_1	S_1	C_2
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

		$a_1 b_1$			
C_1	$a_1 b_1$	00	01	11	10
		m_0	m_1	m_3	m_2
0				1	
1			1	1	1

$$C_1 = a_1 b_1 + C_1 b_1 + C_1 a_1 \quad \rightarrow (2)$$

$$C_1 a_1 + (C_1 \oplus a_1) b_1 \quad \rightarrow (1)$$

Prove (1) = (2)